

Detailed Solutions for TD2 and TD3

M1 Brownian Motion and Mathematical Finance

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Course References and Notation

Throughout, notation follows *Poly cours MB.pdf*. In discrete time we write $\tilde{S}_t^i = S_t^i/S_t^0$, $\tilde{V}_t = V_t/S_t^0$, and $V_t = \langle \theta_t, S_t \rangle$. In the Black–Scholes model we use

$$dS_t = \mu S_t dt + \sigma S_t dB_t, \quad S_t^0 = e^{rt},$$

and the risk-neutral probability $\mathbb{Q}$ under which

$$dS_t = rS_t dt + \sigma S_t dB_t^{\mathbb{Q}}.$$

References such as Poly, Eq. (3.9) or Poly, Thm. 1.3 refer to the course poly.

Part I

TD2: AOA and Discrete-Time Model

1 TD2 Key Techniques

This TD is about the bridge between elementary discrete-time martingale calculus and no-arbitrage pricing in a finite trading grid. The notation is the one of Chapter 1 and Appendix A of the course poly, especially Poly, Def. 1.1–1.4, Poly, Thm. 1.1, Poly, Prop. 1.7, and Poly, Def. A.6.

Key techniques.

- **Conditional expectation.** If X is integrable and Y is $\mathcal{F}_n$ -measurable, then

$$\mathbb{E}[YX \mid \mathcal{F}_n] = Y\mathbb{E}[X \mid \mathcal{F}_n]$$

whenever the products are integrable. If X is independent of $\mathcal{F}_n$, then $\mathbb{E}[X \mid \mathcal{F}_n] = \mathbb{E}[X]$. These two rules are the main tools for random walks, because the new increment is independent of the past while the current value is measurable with respect to the past.

- **Martingales.** A process M is an $\mathcal{F}$ -martingale when it is adapted, integrable, and satisfies

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$

For a centered random walk $X_n = \sum_{k=1}^n \xi_k$, the martingale property comes from $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$. Quadratic martingales are obtained by compensating the predictable drift:

$$X_n^2 - \sum_{k=1}^n \mathbb{E}[\xi_k^2]$$

is an $\mathcal{F}$ -martingale.

- **Martingale transforms.** If M is a martingale and Δ_n is bounded and adapted, then

$$X_n = \sum_{k=1}^n \Delta_{k-1}(M_k - M_{k-1})$$

is again a martingale. This is the discrete stochastic-integral rule: Δ_{k-1} is known before the increment $M_k - M_{k-1}$, so the conditional mean of each new gain is zero.

- **AOA.** In the course convention, an arbitrage opportunity is a self-financing strategy with zero initial capital, nonnegative terminal discounted wealth, and a positive probability of strict gain:

$V_0 = 0$ and $\tilde{V}_T \geq 0$ with $\mathbb{P}(\tilde{V}_T > 0) > 0$, where

see Poly, Def. 1.3. Equivalent definitions allowing $V_0 \leq 0$ are reduced to this one by adding or subtracting a pure risk-free holding.

- **Self-financing strategies.** A strategy is self-financing when changes in wealth come only from price moves:

$$\tilde{V}_{t_k} - \tilde{V}_{t_{k-1}} = \langle \theta_{t_k}, \tilde{S}_{t_k} - \tilde{S}_{t_{k-1}} \rangle.$$

Equivalently, rebalancing at t_{k-1} costs no money in discounted units:

$$\langle \theta_{t_{k-1}}, \tilde{S}_{t_{k-1}} \rangle = \langle \theta_{t_k}, \tilde{S}_{t_{k-1}} \rangle.$$

This is Poly, Rem. 1.1.

- **Forwards with deterministic dividends.** A deterministic dividend is removed from the spot price by subtracting its present value. Thus the time-0 forward delivery price for maturity T is

$$F_0 = \left(S_0 - \sum_j D_j e^{-r\bar{t}_j} \right) e^{rT}$$

when the dividend dates $\bar{t}_j$ are before T .

- **Discrete pricing by backward induction.** If an equivalent probability $\mathbb{Q}$ makes $\tilde{S}$ a martingale, then AOA holds by Poly, Thm. 1.1. If a payoff G is replicable, its price is

$$p_{t_k}^G = S_{t_k}^0 \mathbb{E}_{\mathbb{Q}} \left[\frac{G}{S_T^0} \middle| \mathcal{F}_{t_k} \right],$$

as in Poly, Prop. 1.7. In exercises, it is usually more practical to start from the terminal value

$$p_T^G = G$$

and move one step backward. At a node $(t_{k-1}, S_{t_{k-1}}^1 = s)$ of a binomial tree, the two next stock values at t_k are su and sd . With

$$q = \frac{e^{r\Delta t} - d}{u - d},$$

Poly, Thm. 1.2–1.3 gives the one-step recursion

$$p_{t_{k-1}}^G(s) = e^{-r\Delta t} \left(q p_{t_k}^G(su) + (1 - q) p_{t_k}^G(sd) \right).$$

The risky-asset holding $\theta_{t_k}^1$, held over $(t_{k-1}, t_k]$, is the slope between the two successor values:

$$\theta_{t_k}^1(s) = \frac{p_{t_k}^G(su) - p_{t_k}^G(sd)}{su - sd}.$$

In words: first fill the terminal payoffs, then compute prices one column backward, and at each node compute θ^1 from the two successor prices and stock values.

Exercise I.1 (Random walk martingales). *Restatement.* Let $(\xi_n)_{n \geq 1}$ be i.i.d., centered, and of finite variance. With

$$X_0 = 0, \quad X_n = \sum_{k=1}^n \xi_k, \quad \mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n),$$

show that X is a martingale, identify the compensator of X^2 , and then do the same for the product of two independent centered random walks.

Solution. The result uses only the conditional expectation rules recalled in Poly, App. A.5–A.6. First, X_n is $\mathcal{F}_n$ -measurable and integrable, because

$$\mathbb{E}[|X_n|] \leq \sum_{k=1}^n \mathbb{E}[|\xi_k|] = n\mathbb{E}[|\xi_1|] < \infty,$$

where finite variance implies integrability. Moreover,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[X_n + \xi_{n+1} \mid \mathcal{F}_n] = X_n + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n].$$

Since ξ_{n+1} is independent of $\mathcal{F}_n$ and centered, $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[\xi_{n+1}] = 0$. Hence $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$, so X is an $\mathcal{F}$ -martingale in the sense of Poly, Def. A.6.

For the square, expand one step:

$$X_{n+1}^2 = (X_n + \xi_{n+1})^2 = X_n^2 + 2X_n\xi_{n+1} + \xi_{n+1}^2.$$

Taking conditional expectation with respect to $\mathcal{F}_n$ gives

$$\mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n] = X_n^2 + 2X_n\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[\xi_{n+1}^2 \mid \mathcal{F}_n] = X_n^2 + \mathbb{E}[\xi_{n+1}^2].$$

Thus the predictable drift of X^2 at step $n+1$ is $\mathbb{E}[\xi_{n+1}^2]$. Define

$$a_0 = 0, \quad a_n = \sum_{k=1}^n \mathbb{E}[\xi_k^2].$$

The sequence $(a_n)_{n \geq 0}$ is increasing, and because the ξ_k are i.i.d. it is simply $a_n = n\mathbb{E}[\xi_1^2]$. Then

$$\mathbb{E}[X_{n+1}^2 - a_{n+1} \mid \mathcal{F}_n] = X_n^2 + \mathbb{E}[\xi_{n+1}^2] - a_n - \mathbb{E}[\xi_{n+1}^2] = X_n^2 - a_n.$$

Also $X_n^2 - a_n$ is integrable, so $(X_n^2 - a_n)_{n \geq 0}$ is an $\mathcal{F}$ -martingale. The same calculation also shows uniqueness of the compensator among deterministic sequences starting from 0.

Now let $(\eta_n)_{n \geq 1}$ be another centered i.i.d. finite-variance sequence, independent of the ξ 's, and set

$$Y_0 = 0, \quad Y_n = \sum_{\ell=1}^n \eta_\ell, \quad \mathcal{G}_n = \sigma(\xi_1, \dots, \xi_n, \eta_1, \dots, \eta_n).$$

The product $X_n Y_n$ is integrable by Cauchy–Schwarz:

$$\mathbb{E}[|X_n Y_n|] \leq \mathbb{E}[X_n^2]^{1/2} \mathbb{E}[Y_n^2]^{1/2} < \infty.$$

Since X_n and Y_n are $\mathcal{G}_n$ -measurable,

$$\begin{aligned} \mathbb{E}[X_{n+1} Y_{n+1} \mid \mathcal{G}_n] &= \mathbb{E}[(X_n + \xi_{n+1})(Y_n + \eta_{n+1}) \mid \mathcal{G}_n] \\ &= X_n Y_n + Y_n \mathbb{E}[\xi_{n+1} \mid \mathcal{G}_n] + X_n \mathbb{E}[\eta_{n+1} \mid \mathcal{G}_n] + \mathbb{E}[\xi_{n+1} \eta_{n+1} \mid \mathcal{G}_n]. \end{aligned}$$

The first two new terms vanish because the increments are centered and independent of the past $\mathcal{G}_n$. The last conditional expectation is the constant $\mathbb{E}[\xi_{n+1} \eta_{n+1}]$. Under the stated independence of the two sequences this constant is 0, but it is useful to write the general compensator

$$d_0 = 0, \quad d_n = \sum_{k=1}^n \mathbb{E}[\xi_k \eta_k].$$

Then

$$\mathbb{E}[X_{n+1}Y_{n+1} - d_{n+1} \mid \mathcal{G}_n] = X_nY_n + \mathbb{E}[\xi_{n+1}\eta_{n+1}] - d_n - \mathbb{E}[\xi_{n+1}\eta_{n+1}] = X_nY_n - d_n.$$

Therefore $(X_nY_n - d_n)_{n \geq 0}$ is a $\mathcal{G}$ -martingale. In the exact setting of the exercise, $d_n = 0$ for every n . $\square$

Exercise I.2 (Martingale transform). *Restatement.* Let M be an $\mathcal{F}$ -martingale and let $(\Delta_n)_{n \geq 0}$ be bounded and adapted. Prove that

$$X_0 = 0, \quad X_n = \sum_{k=1}^n \Delta_{k-1}(M_k - M_{k-1})$$

is an $\mathcal{F}$ -martingale.

Solution. This is the discrete version of a stochastic integral with predictable integrand. Since Δ is adapted, Δ_{k-1} is $\mathcal{F}_{k-1}$ -measurable before the martingale increment $M_k - M_{k-1}$ is observed. Since Δ is bounded, say $|\Delta_n| \leq c$ a.s., each summand is integrable:

$$\mathbb{E}[|\Delta_{k-1}(M_k - M_{k-1})|] \leq c(\mathbb{E}[|M_k|] + \mathbb{E}[|M_{k-1}|]) < \infty.$$

Thus X_n is integrable. It is also $\mathcal{F}_n$ -measurable, because every term in the sum up to n is $\mathcal{F}_n$ -measurable.

For the martingale property, write

$$X_{n+1} = X_n + \Delta_n(M_{n+1} - M_n).$$

Here X_n and Δ_n are $\mathcal{F}_n$ -measurable. Therefore

$$\begin{aligned} \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] &= X_n + \Delta_n \mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] \\ &= X_n + \Delta_n(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] - M_n) \\ &= X_n. \end{aligned}$$

The last equality is exactly the martingale property of M , as in Poly, Def. A.6. Hence X is an $\mathcal{F}$ -martingale. $\square$

Exercise I.3 (Equivalent definitions of arbitrage). *Restatement.* Three definitions of arbitrage are proposed. The course definition has $V_0 = 0$, terminal wealth $V_T \geq 0$ a.s., and $\mathbb{P}(V_T > 0) > 0$. The other two allow $V_0 \leq 0$, with either $\mathbb{P}(V_T > 0) > 0$ or the weaker nontriviality condition

$$V_0 \neq 0 \quad \text{or} \quad \mathbb{P}(V_T \neq 0) > 0.$$

Show that the three definitions are equivalent.

Solution. Let the three statements be denoted by (i), (ii), and (iii), in the order given in the exercise. The implications

$$(i) \Rightarrow (ii) \Rightarrow (iii)$$

are immediate. Indeed, $V_0 = 0$ implies $V_0 \leq 0$, and if $\mathbb{P}(V_T > 0) > 0$, then also $\mathbb{P}(V_T \neq 0) > 0$.

It remains to prove that (iii) implies the course definition (i), which is the definition used in Poly, Def. 1.3. Assume that a self-financing value process V satisfies

$$V_0 \leq 0, \quad \mathbb{P}(V_T \geq 0) = 1, \quad (V_0 \neq 0 \text{ or } \mathbb{P}(V_T \neq 0) > 0).$$

There are two cases.

If $V_0 = 0$, then the nontriviality condition gives $\mathbb{P}(V_T \neq 0) > 0$. Since $V_T \geq 0$ a.s., this is the same as $\mathbb{P}(V_T > 0) > 0$. Hence V is already an arbitrage in the sense of (i).

If $V_0 < 0$, add to V a pure risk-free holding with initial value $-V_0 > 0$. More explicitly, if S^0 is the risk-free asset, define

$$W_t = -V_0 \frac{S_t^0}{S_0^0}.$$

The process W is self-financing and starts from $W_0 = -V_0$. The combined value process

$$\widehat{V}_t = V_t + W_t$$

is self-financing by linearity of self-financing strategies (Poly, Prop. 1.1), and it satisfies

$$\widehat{V}_0 = V_0 - V_0 = 0.$$

At maturity,

$$\widehat{V}_T = V_T - V_0 \frac{S_T^0}{S_0^0}.$$

Because $V_T \geq 0$ a.s., $V_0 < 0$, and $S_T^0/S_0^0 > 0$, we have $\widehat{V}_T > 0$ a.s. Thus

$$\mathbb{P}(\widehat{V}_T \geq 0) = 1, \quad \mathbb{P}(\widehat{V}_T > 0) = 1,$$

and $\widehat{V}$ is an arbitrage with zero initial capital. Therefore (iii) implies (i), and all three definitions are equivalent. $\square$

Exercise I.4 (Self-financing criterion). *Restatement.* Show that a strategy θ is self-financing if and only if, for every $k = 2, \dots, N$,

$$\langle \theta_{t_{k-1}}, \widetilde{S}_{t_{k-1}} \rangle = \langle \theta_{t_k}, \widetilde{S}_{t_{k-1}} \rangle.$$

Solution. The statement is Poly, Rem. 1.1, with the proof given in Poly, Chap. 4, Ex. 1.1. Recall first that

$$\widetilde{V}_{t_k} = \langle \theta_{t_k}, \widetilde{S}_{t_k} \rangle$$

and that, by definition, a self-financing strategy satisfies

$$\widetilde{V}_{t_k} - \widetilde{V}_{t_{k-1}} = \langle \theta_{t_k}, \widetilde{S}_{t_k} - \widetilde{S}_{t_{k-1}} \rangle$$

for each trading step. If θ is self-financing, then

$$\begin{aligned} \langle \theta_{t_k}, \widetilde{S}_{t_k} \rangle - \langle \theta_{t_{k-1}}, \widetilde{S}_{t_{k-1}} \rangle &= \langle \theta_{t_k}, \widetilde{S}_{t_k} - \widetilde{S}_{t_{k-1}} \rangle \\ &= \langle \theta_{t_k}, \widetilde{S}_{t_k} \rangle - \langle \theta_{t_k}, \widetilde{S}_{t_{k-1}} \rangle. \end{aligned}$$

Cancelling the common term $\langle \theta_{t_k}, \widetilde{S}_{t_k} \rangle$ gives

$$\langle \theta_{t_{k-1}}, \widetilde{S}_{t_{k-1}} \rangle = \langle \theta_{t_k}, \widetilde{S}_{t_{k-1}} \rangle.$$

This says that the rebalance from $\theta_{t_{k-1}}$ to θ_{t_k} , when evaluated at the old prices $\widetilde{S}_{t_{k-1}}$, requires no external cash.

Conversely, assume the displayed equality holds for all $k = 2, \dots, N$. Then

$$\begin{aligned} \widetilde{V}_{t_k} - \widetilde{V}_{t_{k-1}} &= \langle \theta_{t_k}, \widetilde{S}_{t_k} \rangle - \langle \theta_{t_{k-1}}, \widetilde{S}_{t_{k-1}} \rangle \\ &= \langle \theta_{t_k}, \widetilde{S}_{t_k} \rangle - \langle \theta_{t_k}, \widetilde{S}_{t_{k-1}} \rangle \\ &= \langle \theta_{t_k}, \widetilde{S}_{t_k} - \widetilde{S}_{t_{k-1}} \rangle. \end{aligned}$$

This is precisely the self-financing identity. Hence the criterion is equivalent to self-financing. $\square$

Exercise I.5 (Half risky, half risk-free strategy). *Restatement.* In a one-risky-asset market, an investor always rebalances so that half of the portfolio value is invested in the risky asset and half in the risk-free asset. Find the recursions for the holdings and for the portfolio value.

Solution. Write

$$\theta_{t_k} = (\theta_{t_k}^0, \theta_{t_k}^1)$$

for the holdings chosen at t_{k-1} and held over the interval $(t_{k-1}, t_k]$, as in Poly, Def. 1.1. At time t_{k-1} , before the new holdings are chosen, the wealth is

$$V_{t_{k-1}} = \langle \theta_{t_{k-1}}, S_{t_{k-1}} \rangle = \theta_{t_{k-1}}^0 S_{t_{k-1}}^0 + \theta_{t_{k-1}}^1 S_{t_{k-1}}^1.$$

The rule says that after rebalancing at the old prices $S_{t_{k-1}}$,

$$\theta_{t_k}^0 S_{t_{k-1}}^0 = \frac{1}{2} V_{t_{k-1}}, \quad \theta_{t_k}^1 S_{t_{k-1}}^1 = \frac{1}{2} V_{t_{k-1}}.$$

Therefore

$$\boxed{\theta_{t_k}^0 = \frac{\langle \theta_{t_{k-1}}, S_{t_{k-1}} \rangle}{2 S_{t_{k-1}}^0}, \quad \theta_{t_k}^1 = \frac{\langle \theta_{t_{k-1}}, S_{t_{k-1}} \rangle}{2 S_{t_{k-1}}^1}.}$$

This is exactly the recurrence derived in Poly, Chap. 4, Ex. 1.5.

The value at the next date is obtained by keeping the holdings fixed while prices move:

$$\begin{aligned} V_{t_k} &= \theta_{t_k}^0 S_{t_k}^0 + \theta_{t_k}^1 S_{t_k}^1 \\ &= \frac{V_{t_{k-1}}}{2 S_{t_{k-1}}^0} S_{t_k}^0 + \frac{V_{t_{k-1}}}{2 S_{t_{k-1}}^1} S_{t_k}^1. \end{aligned}$$

Hence the wealth recursion is

$$\boxed{V_{t_k} = V_{t_{k-1}} \left(\frac{1}{2} \frac{S_{t_k}^0}{S_{t_{k-1}}^0} + \frac{1}{2} \frac{S_{t_k}^1}{S_{t_{k-1}}^1} \right).}$$

Equivalently, if $S_{t_k}^0/S_{t_{k-1}}^0 = e^{r(t_k - t_{k-1})}$, then

$$V_{t_k} = V_{t_{k-1}} \left(\frac{1}{2} e^{r(t_k - t_{k-1})} + \frac{1}{2} \frac{S_{t_k}^1}{S_{t_{k-1}}^1} \right).$$

□

Exercise I.6 (Numerical self-financing strategy). *Restatement.* Consider $d = 1$, $r = \ln(1.06)$, $t_0 = 0$, $t_1 = 1$, $t_2 = 2$, $S_{t_0}^0 = 100$, and risky prices

$$S_{t_0}^1 = 100, \quad S_{t_1}^1 = 120, \quad S_{t_2}^1 = 100.$$

A self-financing portfolio has

$$V_{t_0} = 1000, \quad V_{t_1} = 1060, \quad V_{t_2} = 1042.$$

Compute $S_{t_k}^0$, $\tilde{V}_{t_k}$, the holdings, and compare with a pure risk-free investment.

Solution. The risk-free asset evolves according to Poly, Eq. (1.1):

$$S_{t_k}^0 = S_{t_0}^0 e^{r(t_k - t_0)}.$$

Since $e^r = 1.06$,

$$S_{t_0}^0 = 100, \quad S_{t_1}^0 = 106, \quad S_{t_2}^0 = 112.36.$$

Thus the discounted wealths are

$$\tilde{V}_{t_0} = \frac{1000}{100} = 10, \quad \tilde{V}_{t_1} = \frac{1060}{106} = 10, \quad \tilde{V}_{t_2} = \frac{1042}{112.36} \simeq 9.274.$$

The discounted risky prices are

$$\tilde{S}_{t_0}^1 = 1, \quad \tilde{S}_{t_1}^1 = \frac{120}{106} \simeq 1.132, \quad \tilde{S}_{t_2}^1 = \frac{100}{112.36} \simeq 0.890.$$

Because the portfolio is self-financing,

$$\tilde{V}_{t_k} - \tilde{V}_{t_{k-1}} = \theta_{t_k}^1 (\tilde{S}_{t_k}^1 - \tilde{S}_{t_{k-1}}^1),$$

since the discounted risk-free asset is identically 1. For the first period,

$$10 - 10 = \theta_{t_1}^1 \left(\frac{120}{106} - 1 \right),$$

so

$$\theta_{t_1}^1 = 0.$$

The initial wealth identity gives

$$1000 = \theta_{t_1}^0 S_{t_0}^0 + \theta_{t_1}^1 S_{t_0}^1 = 100\theta_{t_1}^0,$$

therefore

$$\theta_{t_1}^0 = 10.$$

For the second period,

$$\theta_{t_2}^1 = \frac{\tilde{V}_{t_2} - \tilde{V}_{t_1}}{\tilde{S}_{t_2}^1 - \tilde{S}_{t_1}^1} = \frac{1042/112.36 - 10}{100/112.36 - 120/106} = 3.$$

Using $V_{t_1} = 1060$, the risk-free holding at t_2 is obtained from

$$1060 = \theta_{t_2}^0 S_{t_1}^0 + \theta_{t_2}^1 S_{t_1}^1 = 106\theta_{t_2}^0 + 3 \cdot 120.$$

Hence

$$\theta_{t_2}^0 = \frac{700}{106} = \frac{350}{53} \simeq 6.604.$$

The terminal check is

$$\theta_{t_2}^0 S_{t_2}^0 + \theta_{t_2}^1 S_{t_2}^1 = \frac{350}{53} \cdot 112.36 + 3 \cdot 100 = 1042.$$

A strategy investing only in the risk-free asset and starting from 1000 would have value

$$1000(1.06)^2 = 1123.60$$

at t_2 . The given investor matches the risk-free account over the first period, then holds 3 shares of the risky asset over the second period. Since the risky asset falls from 120 to 100, the final value is

$$1042 < 1123.60.$$

The strategy therefore underperforms the pure risk-free strategy by 81.60. $\square$

Exercise 1.7 (Forward price with deterministic dividends). *Restatement.* Find the no-arbitrage forward delivery price for an asset S maturing at T when the asset pays deterministic cash dividends before maturity.

Solution. Assume first that there is one deterministic dividend $D > 0$ paid at $\bar{t} \in (0, T)$. A long forward entered at time 0 receives the asset at time T , but does not receive the dividend paid before T . Therefore the correct comparison is the *prepaid forward*: buy the asset today and finance the dividend by borrowing its present value $De^{-r\bar{t}}$. When the dividend is paid at $\bar{t}$, it repays exactly that loan.

Thus the time-0 prepaid forward value is

$$S_0 - De^{-r\bar{t}}.$$

If the ordinary forward delivery price is F_0 , its time-0 value is zero when the present value of paying F_0 at T equals the prepaid forward value:

$$F_0e^{-rT} = S_0 - De^{-r\bar{t}}.$$

Hence

$$F_0 = (S_0 - De^{-r\bar{t}})e^{rT} = S_0e^{rT} - De^{r(T-\bar{t})}.$$

If the delivery price were larger, one could sell the forward and buy the prepaid forward replication; if it were smaller, one could buy the forward and short the replication. This is the standard AOA comparison principle of Poly, Cor. 1.1.

For deterministic dividends $D_1, \dots, D_m$ paid at

$$0 < \bar{t}_1 < \dots < \bar{t}_m < T,$$

the present value of the dividends is additive. The time-0 forward delivery price is therefore

$$F_0 = \left(S_0 - \sum_{j=1}^m D_j e^{-r\bar{t}_j} \right) e^{rT} = S_0 e^{rT} - \sum_{j=1}^m D_j e^{r(T-\bar{t}_j)}.$$

More generally, at an intermediate date $t < T$, only future dividends matter:

$$F_t = \left(S_t - \sum_{\bar{t}_j > t} D_j e^{-r(\bar{t}_j - t)} \right) e^{r(T-t)}.$$

□

Exercise I.8 (No-arbitrage bounds for European calls and puts). For $S, K > 0$ and $t \in [0, T]$, prove the standard lower and upper bounds for a European call $C_t(S, T, K)$ and a European put $P_t(S, T, K)$.

Solution. Let $\tau = T - t$. We use the comparison principle in absence of arbitrage: if one terminal payoff is always dominated by another, then its time t price is dominated as well; see Poly, Prop. 1.2 and Cor. 1.1. The terminal payoff of the call is

$$C_T = (S_T - K)^+.$$

First,

$$0 \leq (S_T - K)^+ \leq S_T.$$

The payoff S_T is obtained by holding one unit of the underlying from t to T , whose price is S at time t . Hence

$$0 \leq C_t(S, T, K) \leq S.$$

Moreover $(x)^+ \geq x$ for every real number x , so

$$(S_T - K)^+ \geq S_T - K.$$

The payoff $S_T - K$ is replicated at time t by buying one unit of the underlying and borrowing $Ke^{-r\tau}$ in the bank account. Its time t value is $S - Ke^{-r\tau}$. Therefore

$$C_t(S, T, K) \geq S - Ke^{-r\tau}.$$

Combining this with $C_t(S, T, K) \geq 0$ gives

$$\boxed{\left(S - Ke^{-r(T-t)}\right)^+ \leq C_t(S, T, K) \leq S}.$$

For the put, the terminal payoff is

$$P_T = (K - S_T)^+.$$

Since $0 \leq (K - S_T)^+ \leq K$, the comparison principle gives

$$0 \leq P_t(S, T, K) \leq Ke^{-r\tau}.$$

Also $(K - S_T)^+ \geq K - S_T$. The payoff $K - S_T$ is replicated by lending $Ke^{-r\tau}$ at time t and short-selling one unit of the underlying, so its time t value is $Ke^{-r\tau} - S$. Hence

$$P_t(S, T, K) \geq Ke^{-r\tau} - S.$$

Together with non-negativity this yields

$$\boxed{\left(S - Ke^{-r(T-t)}\right)^- \leq P_t(S, T, K) \leq Ke^{-r(T-t)}}.$$

These are exactly the bounds stated in Poly, Prop. 1.4. □

Exercise I.9 (Convexity of option prices in the strike). For fixed $S > 0$, show that the maps

$$K \mapsto C_t(S, T, K), \quad K \mapsto P_t(S, T, K)$$

are convex on $(0, \infty)$, for every $t \in [0, T]$.

Solution. Fix $K_1, K_2 > 0$ and $\lambda \in [0, 1]$, and set

$$K_\lambda = \lambda K_1 + (1 - \lambda)K_2.$$

For any terminal value x , the map $K \mapsto (x - K)^+$ is convex. Consequently,

$$(S_T - K_\lambda)^+ \leq \lambda (S_T - K_1)^+ + (1 - \lambda) (S_T - K_2)^+.$$

The right-hand side is the terminal payoff of a portfolio holding λ calls with strike K_1 and $1 - \lambda$ calls with strike K_2 . By the no-arbitrage comparison principle Poly, Cor. 1.1,

$$C_t(S, T, K_\lambda) \leq \lambda C_t(S, T, K_1) + (1 - \lambda)C_t(S, T, K_2).$$

Thus $K \mapsto C_t(S, T, K)$ is convex.

The same proof applies to puts, because $K \mapsto (K - x)^+$ is convex for each fixed x . Therefore

$$(K_\lambda - S_T)^+ \leq \lambda (K_1 - S_T)^+ + (1 - \lambda) (K_2 - S_T)^+,$$

and hence

$$P_t(S, T, K_\lambda) \leq \lambda P_t(S, T, K_1) + (1 - \lambda)P_t(S, T, K_2).$$

This proves the two convexity statements of Poly, Prop. 1.5. □

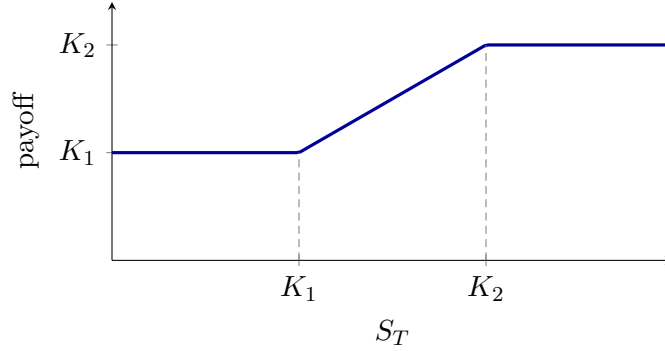
Exercise I.10 (A capped and floored payoff). Consider the derivative with maturity payoff

$$H = \min(\max(S_T, K_1), K_2), \quad K_1 < K_2.$$

Plot the payoff and express its time 0 price using two call prices.

Solution. The payoff is constant at level K_1 below K_1 , equal to the underlying between K_1 and K_2 , and constant at level K_2 above K_2 :

$$H = \begin{cases} K_1, & S_T \leq K_1, \\ S_T, & K_1 < S_T < K_2, \\ K_2, & S_T \geq K_2. \end{cases}$$



Algebraically, for every $x \geq 0$,

$$\min(\max(x, K_1), K_2) = K_1 + (x - K_1)^+ - (x - K_2)^+.$$

Taking $x = S_T$, the derivative is therefore replicated by: holding a zero-coupon bond paying K_1 at T , buying one call with strike K_1 , and selling one call with strike K_2 . By the replication pricing principle Poly, Def. 1.9 and Prop. 1.7,

$$H_0 = K_1 e^{-rT} + C_0(S, T, K_1) - C_0(S, T, K_2).$$

□

Exercise I.11 (Payoff diagrams for elementary option strategies). Plot the terminal payoffs of the listed strategies. The diagrams below are terminal payoffs before taking account of option premiums.

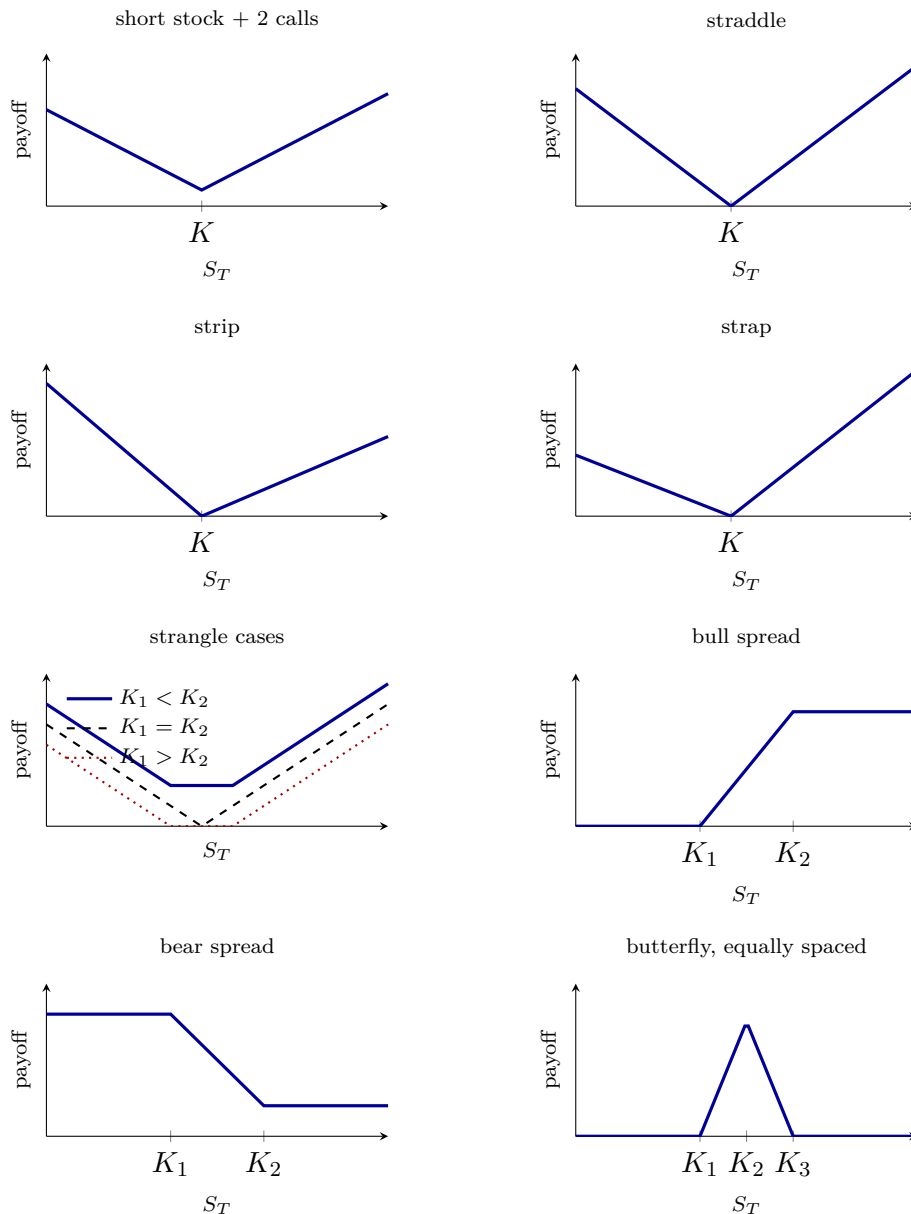
Solution. Write $x = S_T$. The terminal payoff of each strategy is the sum of its component terminal payoffs; option premiums are paid at inception and are therefore not included in the maturity payoff diagrams.

short stock + two calls at K :	$-x + 2(x - K)^+$,
long call + long put at K :	$(x - K)^+ + (K - x)^+ = x - K $,
strip: one call + two puts at K :	$(x - K)^+ + 2(K - x)^+$,
strap: two calls + one put at K :	$2(x - K)^+ + (K - x)^+$,
strangle: call at K_1 + put at K_2 :	$(x - K_1)^+ + (K_2 - x)^+$,
bull spread, $K_1 < K_2$:	$(x - K_1)^+ - (x - K_2)^+$,
bear spread, $K_1 < K_2$:	$(x - K_2)^+ - (x - K_1)^+$,
butterfly, $K_1 < K_2 < K_3$:	$(x - K_1)^+ - 2(x - K_2)^+ + (x - K_3)^+$.

The first strategy is a straddle up to the deterministic cash shift K : by put-call parity at maturity,

$$-x + 2(x - K)^+ + K = (x - K)^+ + (K - x)^+ = |x - K|.$$

The usual long straddle is the second strategy: it pays when the underlying finishes far away from the common strike in either direction.



For the butterfly, the right tail is

$$(x - K_1) - 2(x - K_2) + (x - K_3) = 2K_2 - K_1 - K_3, \quad x \geq K_3.$$

Thus the usual tent-shaped butterfly with zero right tail requires equally spaced strikes, i.e. $K_2 = (K_1 + K_3)/2$. Without this spacing condition the payoff becomes a non-zero constant for $x \geq K_3$. $\square$

Exercise I.12 (Two-step binomial call). In the two-step binomial model with $S_0^1 = 100$, $u = 1.5$, $d = 0.5$, $T = 4$, $N = 2$, $r = -\frac{1}{2} \ln(0.8)$, price the European call with strike $K = 100$, and fill the pricing and hedging tree.

Solution. Here $\Delta t = T/N = 2$, hence

$$e^{r\Delta t} = e^{-\ln(0.8)} = \frac{1}{0.8} = 1.25.$$

By the binomial risk-neutral formula Poly, Thm. 1.2, Eq. (1.8),

$$q = \frac{e^{r\Delta t} - d}{u - d} = \frac{1.25 - 0.5}{1.5 - 0.5} = 0.75.$$

The one-step discount factor is $e^{-r\Delta t} = 0.8$. At maturity the call payoffs are

$$(225 - 100)^+ = 125, \quad (75 - 100)^+ = 0, \quad (25 - 100)^+ = 0.$$

Backward induction Poly, Thm. 1.3, Eq. (1.12) gives

$$p_{t_1}^G(150) = 0.8(0.75 \cdot 125 + 0.25 \cdot 0) = 75,$$

$$p_{t_1}^G(50) = 0,$$

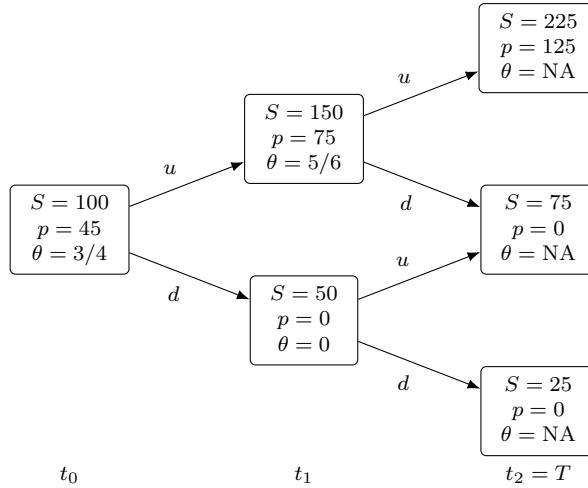
$$p_{t_0}^G(100) = 0.8(0.75 \cdot 75 + 0.25 \cdot 0) = 45.$$

The hedge ratios are the binomial deltas from Poly, Thm. 1.3, Eq. (1.13):

$$\theta_{t_1}^1(100) = \frac{75 - 0}{150 - 50} = \frac{3}{4},$$

$$\theta_{t_2}^1(150) = \frac{125 - 0}{225 - 75} = \frac{5}{6},$$

$$\theta_{t_2}^1(50) = \frac{0 - 0}{75 - 25} = 0.$$



Thus the time 0 no-arbitrage price of the call is 45, and the missing tree entries are those displayed above. $\square$

Exercise I.13 (Three-step binomial tree). The statement calls the model trinomial, but the displayed data form a recombining binomial tree with $u = 1.5$ and $d = 0.5$. For $S_0^1 = 80$, $N = T = 3$, $r = \ln(1.1)$, and $K = 80$, price the European call and describe the hedge along the scenarios

$$80 \rightarrow 120 \rightarrow 60 \rightarrow 90, \quad 80 \rightarrow 120 \rightarrow 60 \rightarrow 30.$$

Solution. Here $\Delta t = 1$ and $e^{r\Delta t} = 1.1$. The risk-neutral probability is

$$q = \frac{e^{r\Delta t} - d}{u - d} = \frac{1.1 - 0.5}{1.5 - 0.5} = 0.6,$$

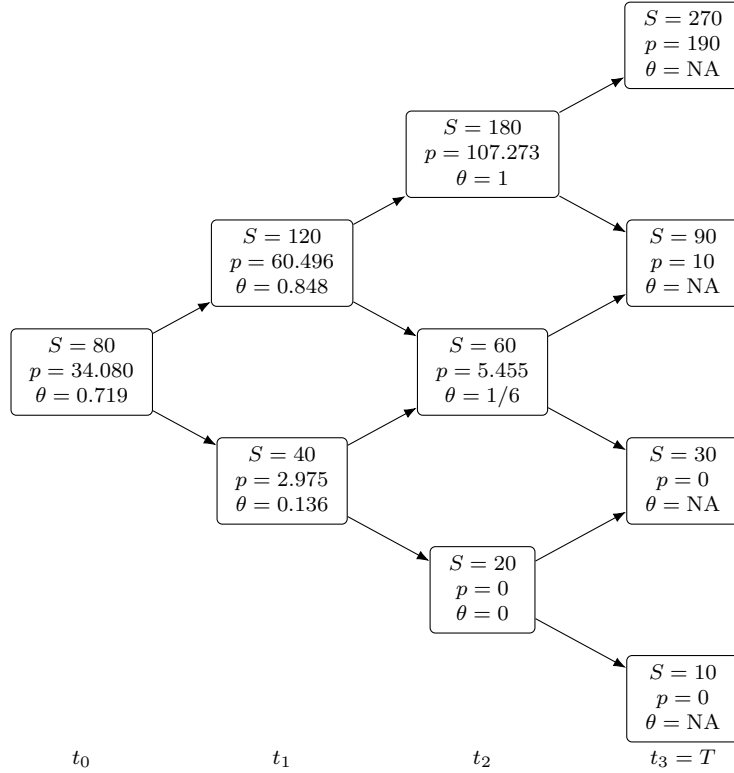
as in Poly, Thm. 1.2, Eq. (1.8). The one-step discount factor is $1/1.1$. Terminal payoffs are

$$(270 - 80)^+ = 190, \quad (90 - 80)^+ = 10, \quad (30 - 80)^+ = 0, \quad (10 - 80)^+ = 0.$$

Backward induction gives

date and node	p^G	θ^1 for the next step
$t_2, S = 180$	107.273	1
$t_2, S = 60$	5.455	1/6
$t_2, S = 20$	0	0
$t_1, S = 120$	60.496	0.848
$t_1, S = 40$	2.975	0.136
$t_0, S = 80$	34.080	0.719

where each θ^1 is computed as the slope of the option value over the two successor stock values, as in Poly, Thm. 1.3, Eq. (1.13). The numbers are rounded only after exact arithmetic with $r = \ln(1.1)$ and the verified value $q = 0.6$.



We now follow the hedge along the scenario $80 \rightarrow 120 \rightarrow 60$. At t_0 , the writer receives 34.080, buys

$$\theta_{t_1}^1 = 0.719$$

shares, at cost $0.719 \cdot 80 = 57.521$, and borrows

$$57.521 - 34.080 = 23.441.$$

At t_1 , after the up move to $S = 120$, the debt has grown to $23.441 \cdot 1.1 = 25.785$, while the shares are worth 86.281. The net portfolio value is $60.496 = p_{t_1}^G(120)$. The new hedge ratio is 0.848, so the writer buys $0.848 - 0.719 = 0.129$ additional shares. The new debt is

$$0.848 \cdot 120 - 60.496 = 41.322.$$

At t_2 , after the down move to $S = 60$, this debt has grown to $41.322 \cdot 1.1 = 45.455$, while the 0.848 shares are worth 50.909. The net value is $5.455 = p_{t_2}^G(60)$. The new hedge ratio is $1/6$, so the writer sells $0.848 - 1/6 = 0.682$ shares and reduces the debt to

$$\frac{1}{6} \cdot 60 - 5.455 = 4.545.$$

If the last move is up to 90, the remaining shares are worth 15 and the debt has grown to 5, so the portfolio value is $10 = (90 - 80)^+$. If the last move is down to 30, the shares are worth 5 and the debt is 5, so the portfolio value is $0 = (30 - 80)^+$. The same hedge at t_2 therefore replicates the call in both requested continuations. $\square$

Exercise I.14 (Non-uniqueness of risk-neutral probabilities in a trinomial tree). Show that in a trinomial one-period model the risk-neutral probability need not be unique.

Solution. In the binomial model, the two unknown probabilities are pinned down by two linear equations, which is why Poly, Thm. 1.2 gives a unique risk-neutral probability. In a trinomial one-step model, suppose the risky return is one of

$$d < m < u$$

and write

$$p_d = \mathbb{Q}(Y = d), \quad p_m = \mathbb{Q}(Y = m), \quad p_u = \mathbb{Q}(Y = u).$$

The risk-neutral martingale condition for the discounted stock is

$$\mathbb{E}^{\mathbb{Q}}[Y] = e^{r\Delta t}.$$

Thus the probabilities must solve

$$p_d + p_m + p_u = 1, \quad dp_d + mp_m + up_u = e^{r\Delta t}.$$

This is a system of two equations in three unknowns. If $a = e^{r\Delta t}$, then for any chosen value $\lambda = p_m$,

$$p_u = \frac{a - d - \lambda(m - d)}{u - d}, \quad p_d = \frac{u - a - \lambda(u - m)}{u - d}.$$

For an interval of values of λ , these three numbers are non-negative and sum to 1. Hence there is generally a whole family of risk-neutral probabilities.

For example, take

$$d = 0.8, \quad m = 1, \quad u = 1.2, \quad e^{r\Delta t} = 1.05.$$

Then

$$p_u = 0.625 - 0.5\lambda, \quad p_d = 0.375 - 0.5\lambda, \quad p_m = \lambda.$$

Every $\lambda \in [0, 0.75]$ gives a valid risk-neutral probability. The risk-neutral probability is therefore not unique. In financial terms, the single risky asset and the money-market account do not span all three states of the world in one step, so the one-step trinomial market is typically incomplete; compare with Poly, Def. 1.7. $\square$

Exercise I.15 (Arrow–Debreu prices and the risk-neutral probability). In a complete discrete-time market with one risky asset and a risk-neutral probability, assume $S_{t_k}^1$ takes finitely many values. Compare the prices of Arrow–Debreu claims with the risk-neutral probabilities, and generalize to $d > 1$.

Solution. Let the possible terminal values of the risky asset be

$$s_1, \dots, s_M.$$

For each j , define the Arrow–Debreu payoff

$$G^j = \mathbf{1}_{\{S_T^1 = s_j\}}.$$

Because the market is complete, G^j is replicable. By the risk-neutral pricing formula for a replicable contingent claim Poly, Prop. 1.7,

$$p_{t_k}^{G^j} = e^{-r(T-t_k)} \mathbb{E}^{\mathbb{Q}} \left[\mathbf{1}_{\{S_T^1 = s_j\}} \mid \mathcal{F}_{t_k} \right] = e^{-r(T-t_k)} \mathbb{Q} \left(S_T^1 = s_j \mid \mathcal{F}_{t_k} \right).$$

Equivalently, the T -capitalized price is the conditional risk-neutral probability:

$$e^{r(T-t_k)} p_{t_k}^{G^j} = \mathbb{Q} \left(S_T^1 = s_j \mid \mathcal{F}_{t_k} \right).$$

At the initial date, when $\mathcal{F}_{t_0}$ is trivial, this becomes

$$p_{t_0}^{G^j} = e^{-rT} \mathbb{Q}(S_T^1 = s_j).$$

Thus Arrow–Debreu prices are exactly discounted state probabilities under $\mathbb{Q}$. Knowing all Arrow–Debreu prices is the same as knowing the risk-neutral law of S_T^1 .

For $d > 1$, let the vector of risky terminal prices take values

$$z_\ell = (z_\ell^1, \dots, z_\ell^d), \quad \ell = 1, \dots, L.$$

The Arrow–Debreu payoff for state z_ℓ is

$$G^\ell = \mathbf{1}_{\{(S_T^1, \dots, S_T^d) = z_\ell\}}.$$

The same argument gives

$$p_{t_k}^{G^\ell} = e^{-r(T-t_k)} \mathbb{Q} \left((S_T^1, \dots, S_T^d) = z_\ell \mid \mathcal{F}_{t_k} \right).$$

Consequently the family of Arrow–Debreu prices recovers the full joint risk-neutral distribution of the terminal risky-asset vector. This is the finite-state version of the replication and valuation principle developed in Poly, Sec. 1.4. $\square$

Part II

TD3: The Black–Scholes Model

2 TD3 Key Techniques

Key techniques.

- **Ito processes and differential rules.** Most continuous-time calculations in this TD reduce to the following template. If

$$X_t = X_0 + \int_0^t a_s \, ds + \int_0^t b_s \, dB_s, \quad \text{so} \quad dX_t = a_t \, dt + b_t \, dB_t,$$

then a_t is the drift coefficient, b_t is the volatility coefficient, the finite-variation part is $a_t \, dt$, and the martingale part is $b_t \, dB_t$. The symbolic multiplication table is

$$(dB_t)^2 = dt, \quad dt \, dB_t = 0, \quad (dt)^2 = 0.$$

It is a shorthand for the quadratic-variation identities in Poly, Ch. 2.5.

- **Quadratic variation and covariation.** For Ito processes

$$dX_t = a_t \, dt + b_t \, dB_t, \quad dY_t = \alpha_t \, dt + \beta_t \, dB_t,$$

the quadratic variation and covariation are

$$d\langle X \rangle_t = b_t^2 \, dt, \quad d\langle X, Y \rangle_t = b_t \beta_t \, dt.$$

Finite-variation terms do not contribute. In particular

$$d\langle B \rangle_t = dt, \quad d\langle t, X \rangle_t = 0.$$

For the BMS stock $dS_t = \mu S_t \, dt + \sigma S_t \, dB_t$,

$$d\langle S \rangle_t = \sigma^2 S_t^2 \, dt, \quad d\langle \tilde{S} \rangle_t = \sigma^2 \tilde{S}_t^2 \, dt.$$

- **Ito formula.** If $f \in C^{1,2}$ and X is as above, then

$$df(t, X_t) = \left(\partial_t f + a_t \partial_x f + \frac{1}{2} b_t^2 \partial_{xx} f \right) (t, X_t) \, dt + b_t \partial_x f(t, X_t) \, dB_t.$$

For two Ito processes driven by the same Brownian motion, the two-dimensional version adds the mixed covariation term:

$$\begin{aligned} df(t, X_t, Y_t) &= \partial_t f \, dt + \partial_x f \, dX_t + \partial_y f \, dY_t \\ &\quad + \frac{1}{2} \partial_{xx} f \, d\langle X \rangle_t + \partial_{xy} f \, d\langle X, Y \rangle_t + \frac{1}{2} \partial_{yy} f \, d\langle Y \rangle_t. \end{aligned}$$

This is the main tool for deriving lognormal laws, BMS PDEs, Greeks, and hedge dynamics; see Poly, Thm. 2.5.

- **Ito integration by parts.** For two Ito processes,

$$d(X_t Y_t) = X_t \, dY_t + Y_t \, dX_t + d\langle X, Y \rangle_t.$$

Equivalently,

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s \, dY_s + \int_0^t Y_s \, dX_s + \langle X, Y \rangle_t.$$

If one factor has finite variation, the covariation term is zero. This is useful when discounting: for $D_t = e^{-rt}$,

$$d(D_t S_t) = D_t \, dS_t + S_t \, dD_t$$

because $\langle D, S \rangle = 0$. See Poly, Prop. 2.26.

- **Stochastic integrals and martingales.** If H is adapted and square-integrable, then

$$M_t = \int_0^t H_s \, dB_s$$

is a martingale with

$$\mathbb{E}[M_t] = 0, \quad \mathbb{E}[M_t^2] = \mathbb{E} \left[\int_0^t H_s^2 \, ds \right], \quad \langle M \rangle_t = \int_0^t H_s^2 \, ds.$$

In finance, this is why discounted self-financing wealth is a martingale under a risk-neutral measure, once integrability/admissibility is checked.

- **Black–Merton–Scholes dynamics.** The risky asset satisfies

$$dS_t = \mu S_t \, dt + \sigma S_t \, dB_t, \quad S_t^0 = e^{rt},$$

hence

$$S_t = S_0 \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma B_t \right), \quad d\tilde{S}_t = (\mu - r) \tilde{S}_t \, dt + \sigma \tilde{S}_t \, dB_t.$$

This follows from applying Ito formula to $\log S_t$, where the correction $-\sigma^2/2$ comes from $d\langle S \rangle_t = \sigma^2 S_t^2 \, dt$. The discounted dynamics are obtained by integration by parts:

$$d\tilde{S}_t = d(e^{-rt} S_t) = (\mu - r) \tilde{S}_t \, dt + \sigma \tilde{S}_t \, dB_t.$$

This is the model and notation of Poly, Eqs. (3.2)–(3.3).

- **Self-financing wealth.** If $(x, \theta) \in \mathbb{R} \times \mathcal{A}$ is self-financing and $S_0^0 = 1$, then

$$\tilde{V}_t^{x, \theta} = x + \int_0^t \theta_u \, d\tilde{S}_u = x + \int_0^t (\mu - r) \theta_u \tilde{S}_u \, du + \int_0^t \sigma \theta_u \tilde{S}_u \, dB_u,$$

as in Poly, Eq. (3.8). Under a risk-neutral probability this becomes a martingale stochastic integral.

- **Girsanov and the market price of risk.** With $\lambda = (\mu - r)/\sigma$, define

$$Z_t = \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left(-\lambda B_t - \frac{1}{2} \lambda^2 t \right).$$

The square on λ is essential. By Girsanov $B_t^{\mathbb{Q}} = B_t + \lambda t$ is a Brownian motion under $\mathbb{Q}$, and

$$dS_t = r S_t \, dt + \sigma S_t \, dB_t^{\mathbb{Q}}, \quad d\tilde{S}_t = \sigma \tilde{S}_t \, dB_t^{\mathbb{Q}}.$$

Thus the physical drift is μ under $\mathbb{P}$, while the risk-neutral drift is r under $\mathbb{Q}$. See Poly, Eq. (3.6), Prop. 3.1.

- **How the BMS PDE appears.** Let $p_t = u(t, S_t)$ be the candidate price of a payoff $\Phi(S_T)$. Under $\mathbb{Q}$,

$$dS_t = r S_t \, dt + \sigma S_t \, dB_t^{\mathbb{Q}}.$$

Ito formula gives

$$du(t, S_t) = \left(\partial_t u + r S_t \partial_x u + \frac{1}{2} \sigma^2 S_t^2 \partial_{xx} u \right) dt + \sigma S_t \partial_x u \, dB_t^{\mathbb{Q}}.$$

Discounting with integration by parts,

$$\begin{aligned} d(e^{-rt}u(t, S_t)) &= e^{-rt} \left(\partial_t u + rS_t \partial_x u + \frac{1}{2} \sigma^2 S_t^2 \partial_{xx} u - ru \right) dt \\ &\quad + e^{-rt} \sigma S_t \partial_x u \, dB_t^{\mathbb{Q}}. \end{aligned}$$

A discounted price must be a $\mathbb{Q}$ -martingale, so the drift must vanish. This gives the BMS PDE

$$\partial_t u + rx \partial_x u + \frac{1}{2} \sigma^2 x^2 \partial_{xx} u = ru.$$

- **Risk-neutral pricing.** For a square-integrable payoff G at maturity T ,

$$p_t^G = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} G \mid \mathcal{F}_t \right],$$

which is Poly, Eq. (3.9). Completeness of the one-dimensional Brownian model gives the corresponding replicating strategy.

- **Lognormal laws.** For $\tau = T - t$, conditionally on $S_t = x$,

$$S_T = x \exp \left(\left(r - \frac{\sigma^2}{2} \right) \tau + \sigma \sqrt{\tau} Z \right) \quad \text{under } \mathbb{Q},$$

where $Z \sim \mathcal{N}(0, 1)$. Under $\mathbb{P}$, replace r by μ . Many probability questions in TD3 are just threshold conversions. For example, under a drift $m \in \{\mu, r\}$,

$$\mathbb{P}_m(S_T \geq K \mid S_t = x) = N \left(\frac{\log(x/K) + (m - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \right).$$

- **Black–Merton–Scholes formula and PDE.** For a European call with strike K and maturity T ,

$$C(t, x) = xN(d_1) - Ke^{-r\tau}N(d_2),$$

where

$$d_1 = \frac{\log(x/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

The price u solves the BMS PDE

$$\partial_t u + rx \partial_x u + \frac{1}{2} \sigma^2 x^2 \partial_{xx} u = ru, \quad u(T, x) = \Phi(x),$$

as in Poly, Eqs. (3.12), (3.17)–(3.18). The usual proof is: write the conditional lognormal expectation under $\mathbb{Q}$, split the event $S_T > K$, and use the Gaussian identity $x\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ when differentiating.

- **Greeks and delta hedging.** For the call,

$$\Delta = \partial_x C = N(d_1), \quad \Gamma = \partial_{xx} C = \frac{\varphi(d_1)}{x\sigma\sqrt{\tau}},$$

and

$$\Theta = \partial_t C = -\frac{x\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2).$$

The replicating risky holding is the delta: $\theta_t = \partial_x u(t, S_t)$, cf. Poly, Eq. (3.16). The reason is visible in the Ito calculation above:

$$d(e^{-rt}u(t, S_t)) = \partial_x u(t, S_t) \, d\tilde{S}_t$$

when u solves the PDE. Thus the stochastic part of the discounted price is matched by holding $\partial_x u(t, S_t)$ shares of the risky asset.

- **Digital options.** A digital call $G = \mathbf{1}_{\{S_T \geq K\}}$ has price

$$C^D(t, x, K) = e^{-r\tau} N(d_2), \quad \partial_x C^D(t, x, K) = \frac{e^{-r\tau} \varphi(d_2)}{x\sigma\sqrt{\tau}}.$$

It is also the strike derivative of the European call: $C^D(t, x, K) = -\partial_K C^E(t, x, K)$. This identity is often proved without computing the BMS formula: compare a call spread of width ΔK with a digital payoff, then pass to the limit. The replicating risky holding is $\partial_x C^D(t, S_t, K)$, provided $t < T$.

- **Time-dependent deterministic volatility.** If $dS_t/S_t = \mu dt + \sigma_t dB_t$, set

$$I_t = \int_t^T \sigma_s^2 ds, \quad \Sigma_t = \sqrt{I_t}.$$

Then the call formula remains

$$C(t, x) = xN(d_1(t, x)) - Ke^{-r(T-t)}N(d_2(t, x)),$$

with

$$d_1(t, x) = \frac{\log(x/K) + r(T-t) + \frac{1}{2}I_t}{\Sigma_t}, \quad d_2(t, x) = d_1(t, x) - \Sigma_t.$$

At $t = 0$, this equals the constant-volatility call with $\bar{\sigma}^2 = T^{-1} \int_0^T \sigma_s^2 ds$. The important point is that only the future integrated variance $I_t = \int_t^T \sigma_s^2 ds$ enters the conditional lognormal law. Unless σ_t is constant, this cannot be represented on every subinterval by one fixed constant volatility for all t .

- **Path-dependent payoffs and Jensen arguments.** For an Asian call,

$$G = \left(\frac{1}{T} \int_0^T S_t dt - K \right)^+,$$

convexity of $x \mapsto (x - K)^+$, Fubini, and risk-neutral pricing give a comparison with the European call. The standard move is to bound the payoff pathwise or after expectation by averaging European call payoffs, then use that $t \mapsto e^{-rt}S_t$ is a $\mathbb{Q}$ -martingale.

- **Compound calls.** A call on a call is priced by conditioning first to T_1 , then to T . The threshold x_1 solves $C(T_1, x_1, T, K) = K_1$. With $\tau = T_1 - t$ and d chosen so that $S_{T_1} > x_1$ is exactly $Z > -d$, the final expression is written with a bivariate normal probability. The method is more important than the final formula: introduce independent Gaussian increments for $[t, T_1]$ and $[T_1, T]$, convert each exercise event into a linear inequality in the Gaussian variables, then evaluate the resulting two-dimensional normal probability.

Exercise II.1 (Self-financing wealth and the risk-neutral measure). **Restatement.** In the Black–Scholes model, express the discounted value of a self-financing strategy (x, θ) , compute its first two moments, and show that the Girsanov measure with market price of risk $(\mu - r)/\sigma$ is risk-neutral.

Solution. We use the convention $S_0^0 = 1$, so that initial capital x is also initial discounted capital. By self-financing, Poly, Eq. (3.8) gives

$$\tilde{V}_t^{x, \theta} = x + \int_0^t \theta_u d\tilde{S}_u = x + \int_0^t (\mu - r)\theta_u \tilde{S}_u du + \int_0^t \sigma \theta_u \tilde{S}_u dB_u.$$

If $S_0^0 \neq 1$, replace x by x/S_0^0 in this formula.

Set

$$A_t = (\mu - r) \int_0^t \theta_u \tilde{S}_u \, du, \quad M_t = \sigma \int_0^t \theta_u \tilde{S}_u \, dB_u.$$

Under the usual square-integrability assumptions, M is a martingale with $\mathbb{E}[M_t] = 0$, hence

$$\mathbb{E}[\tilde{V}_t^{x,\theta}] = x + \mathbb{E}[A_t] = x + \mathbb{E} \left[\int_0^t (\mu - r) \theta_u \tilde{S}_u \, du \right].$$

The variance is

$$\text{Var}(\tilde{V}_t^{x,\theta}) = \text{Var}(A_t + M_t) = \text{Var}(A_t) + \mathbb{E} \left[\int_0^t \sigma^2 \theta_u^2 \tilde{S}_u^2 \, du \right] + 2 \, \text{Cov}(A_t, M_t).$$

The covariance term is kept in the general adapted case; it vanishes only under extra assumptions, for instance for deterministic integrands.

Now put

$$\lambda = \frac{\mu - r}{\sigma}, \quad Z_t = \exp \left(-\lambda B_t - \frac{1}{2} \lambda^2 t \right).$$

The density must contain λ^2 in the deterministic term. Since λ is constant, Novikov's condition is immediate and $d\mathbb{Q}/d\mathbb{P}|_{\mathcal{F}_t} = Z_t$ defines a probability measure equivalent to $\mathbb{P}$. By Girsanov's theorem, stated in the poly before the BMS chapter and used in Poly, Eq. (3.6), the process

$$B_t^{\mathbb{Q}} = B_t + \lambda t$$

is a Brownian motion under $\mathbb{Q}$. Substituting $dB_t = dB_t^{\mathbb{Q}} - \lambda dt$ in $dS_t = \mu S_t dt + \sigma S_t dB_t$ gives

$$dS_t = (\mu - \sigma\lambda) S_t dt + \sigma S_t dB_t^{\mathbb{Q}} = r S_t dt + \sigma S_t dB_t^{\mathbb{Q}}.$$

Therefore

$$d\tilde{S}_t = \sigma \tilde{S}_t dB_t^{\mathbb{Q}},$$

so the discounted risky asset is a $\mathbb{Q}$ -martingale. Thus $\mathbb{Q}$ is a risk-neutral probability measure. $\square$

Exercise II.2 (European call, Greeks, and replication). **Restatement.** For $G = (S_T - K)^+$, derive the BMS price, the exercise probabilities under $\mathbb{P}$ and $\mathbb{Q}$, the Greeks, the PDE, and the delta-hedging replication.

Solution. Let $\tau = T - t$. Under the risk-neutral measure of Exercise 1,

$$S_T = x \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma \sqrt{\tau} Z \right), \quad Z \sim \mathcal{N}(0, 1),$$

conditionally on $S_t = x$. Therefore, using Poly, Eq. (3.9),

$$u(t, x) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r\tau} (S_T - K)^+ \mid S_t = x \right] = x N(d_1) - K e^{-r\tau} N(d_2),$$

where

$$d_1 = \frac{\log(x/K) + (r + \frac{1}{2} \sigma^2) \tau}{\sigma \sqrt{\tau}}, \quad d_2 = d_1 - \sigma \sqrt{\tau} = \frac{\log(x/K) + (r - \frac{1}{2} \sigma^2) \tau}{\sigma \sqrt{\tau}}.$$

This is the Black–Merton–Scholes formula of Poly, Eq. (3.12).

At time 0, the option is exercised iff $S_T \geq K$. Under $\mathbb{P}$,

$$\mathbb{P}(S_T \geq K) = N \left(\frac{\log(S_0/K) + (\mu - \frac{1}{2} \sigma^2) T}{\sigma \sqrt{T}} \right),$$

whereas under $\mathbb{Q}$,

$$\mathbb{Q}(S_T \geq K) = N\left(\frac{\log(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right).$$

The only change is the drift: μ under $\mathbb{P}$, r under $\mathbb{Q}$.

On $[0, T) \times \mathbb{R}_+^*$, the functions d_1, d_2 are $C^{1,2}$, and so is u . The useful identity

$$x\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$$

follows by comparing the Gaussian densities. Differentiating the call formula then gives

$$\begin{aligned}\Delta(t, x) &= \partial_x u(t, x) = N(d_1), \\ \Gamma(t, x) &= \partial_{xx} u(t, x) = \frac{\varphi(d_1)}{x\sigma\sqrt{\tau}},\end{aligned}$$

and

$$\Theta(t, x) = \partial_t u(t, x) = -\frac{x\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2).$$

Substituting these expressions gives

$$\partial_t u(t, x) + rx\partial_x u(t, x) + \frac{1}{2}\sigma^2 x^2 \partial_{xx} u(t, x) = ru(t, x),$$

with terminal condition $u(T, x) = (x - K)^+$. This is the BMS PDE Poly, Eq. (3.17) with the call payoff.

Define the risky holding

$$\vartheta_t^* = \partial_x u(t, S_t) = N(d_1(t, S_t)), \quad 0 \leq t < T.$$

It is adapted and bounded by 1. Since $\mathbb{E}^\mathbb{Q}[\int_0^T \tilde{S}_t^2 dt] < \infty$ in the lognormal model, $\vartheta^* \in \mathcal{A}$.

Finally apply Ito's formula under $\mathbb{Q}$ to $e^{-rt}u(t, S_t)$. The PDE cancels the drift and leaves

$$d(e^{-rt}u(t, S_t)) = e^{-rt}\partial_x u(t, S_t)\sigma S_t dB_t^\mathbb{Q} = \vartheta_t^* d\tilde{S}_t.$$

Thus the self-financing strategy with initial capital $u(0, S_0)$ and risky holding ϑ^* has discounted value $e^{-rt}u(t, S_t)$, and at T its value is $(S_T - K)^+$. It replicates the call, so the no-arbitrage price is $p_t = u(t, S_t)$. $\square$

Exercise II.3 (A constrained strategy cannot replicate nonnegative claims). **Restatement.** Let $G \geq 0$ with $\mathbb{E}^\mathbb{Q}[G^2] < \infty$. Show that G cannot be replicated by a strategy constrained to hold $\theta_t = c/S_t$ units of the risky asset, with $c \in \mathbb{R}^*$.

Solution. Assume, toward a contradiction, that a self-financing strategy (x, θ) with $\theta_t = c/S_t$ replicates G . In discounted form, under the risk-neutral measure,

$$\tilde{V}_T^{x, \theta} = x + \int_0^T \theta_t d\tilde{S}_t = x + \int_0^T \frac{c}{S_t} \sigma \tilde{S}_t dB_t^\mathbb{Q} = x + c\sigma \int_0^T e^{-rt} dB_t^\mathbb{Q}.$$

Since $c \neq 0$, the last random variable is Gaussian with mean x and strictly positive variance

$$c^2\sigma^2 \int_0^T e^{-2rt} dt > 0.$$

Hence

$$\mathbb{Q}\left(x + c\sigma \int_0^T e^{-rt} dB_t^\mathbb{Q} < 0\right) > 0.$$

Equivalence of $\mathbb{P}$ and $\mathbb{Q}$ gives the same positivity under $\mathbb{P}$. But a replicating portfolio would satisfy

$$\tilde{V}_T^{x, \theta} = e^{-rT}G \geq 0 \quad \text{almost surely,}$$

which is impossible. Therefore no such constrained strategy can replicate the nonnegative claim. $\square$

Exercise II.4 (Deterministic time-dependent volatility). **Restatement.** Price a European call when $dS_t/S_t = \mu dt + \sigma_t dB_t$, where σ_t is deterministic, positive, and square-integrable. Compute Δ and Θ , and compare with the constant-volatility BMS surface.

Solution. Set

$$I_t = \int_t^T \sigma_s^2 ds, \quad \Sigma_t = \sqrt{I_t}, \quad \tau = T - t.$$

Since $\sigma_t > 0$, the market price of risk is the deterministic function $\lambda_t = (\mu - r)/\sigma_t$. The Girsanov density is

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = \exp \left(- \int_0^T \lambda_s dB_s - \frac{1}{2} \int_0^T \lambda_s^2 ds \right),$$

and $B_t^{\mathbb{Q}} = B_t + \int_0^t \lambda_s ds$ is Brownian under $\mathbb{Q}$. Thus

$$\frac{dS_t}{S_t} = r dt + \sigma_t dB_t^{\mathbb{Q}}.$$

Conditionally on $S_t = x$,

$$S_T = x \exp \left(r\tau - \frac{1}{2} I_t + \int_t^T \sigma_s dB_s^{\mathbb{Q}} \right),$$

and the stochastic integral is Gaussian with mean 0 and variance I_t . Consequently

$$u(t, x) = xN(d_1(t, x)) - Ke^{-r\tau}N(d_2(t, x)),$$

where

$$d_1(t, x) = \frac{\log(x/K) + r\tau + \frac{1}{2}I_t}{\Sigma_t}, \quad d_2(t, x) = d_1(t, x) - \Sigma_t.$$

At $t = 0$, this is exactly the ordinary BMS price with constant volatility

$$\bar{\sigma} = \sqrt{\frac{1}{T} \int_0^T \sigma_s^2 ds},$$

because the total variance to maturity is $\bar{\sigma}^2 T = I_0$.

As in the constant-volatility case,

$$\Delta(t, x) = \partial_x u(t, x) = N(d_1(t, x)),$$

and

$$\Gamma(t, x) = \frac{\varphi(d_1(t, x))}{x\Sigma_t}.$$

The time-dependent BMS PDE is

$$\partial_t u + rx\partial_x u + \frac{1}{2}\sigma_t^2 x^2 \partial_{xx} u = ru.$$

Therefore

$$\Theta(t, x) = \partial_t u(t, x) = -rKe^{-r\tau}N(d_2(t, x)) - \frac{\sigma_t^2 x \varphi(d_1(t, x))}{2\Sigma_t}.$$

Suppose that the whole price surface on $[0, T]$ were the price surface of a constant-volatility model with some volatility $\hat{\sigma}$. Then the total remaining variance would have to agree for every t :

$$\hat{\sigma}^2(T - t) = \int_t^T \sigma_s^2 ds.$$

Differentiating in t gives $\hat{\sigma}^2 = \sigma_t^2$ for almost every t . Hence the only way a constant-volatility surface can match for all dates is that σ_t is constant almost everywhere. If σ_t is not constant, the match at $t = 0$ does not extend to the whole interval. $\square$

Exercise II.5 (Asian call versus European call). **Restatement.** For the Asian payoff

$$G = \left(\frac{1}{T} \int_0^T S_t \, dt - K \right)^+,$$

show that its time-0 price is no larger than the European call price with the same strike and maturity.

Solution. By risk-neutral pricing, Poly, Eq. (3.9),

$$p_0^{\text{Asian}} = \mathbb{E}^{\mathbb{Q}} \left[e^{-rT} \left(\frac{1}{T} \int_0^T S_t \, dt - K \right)^+ \right].$$

The function $y \mapsto (y - K)^+$ is convex, so Jensen's inequality gives the pathwise bound

$$\left(\frac{1}{T} \int_0^T S_t \, dt - K \right)^+ \leq \frac{1}{T} \int_0^T (S_t - K)^+ \, dt.$$

Thus

$$p_0^{\text{Asian}} \leq \frac{1}{T} \int_0^T e^{-rt} \mathbb{E}^{\mathbb{Q}}[(S_t - K)^+] \, dt.$$

Since $r \geq 0$ in the BMS chapter and S is a $\mathbb{Q}$ -submartingale, the increasing convex function $x \mapsto (x - K)^+$ also yields a submartingale: for $t \leq T$,

$$\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid \mathcal{F}_t] \geq \left(\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] - K \right)^+ = \left(e^{r(T-t)} S_t - K \right)^+ \geq (S_t - K)^+.$$

Taking expectations,

$$\mathbb{E}^{\mathbb{Q}}[(S_t - K)^+] \leq \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

Therefore

$$p_0^{\text{Asian}} \leq e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = p_0^{\text{European}}.$$

□

Exercise II.6 (Digital call as a strike derivative). **Restatement.** Compare a European call with portfolios made of digital calls and nearby European calls. Deduce $C^D = -\partial_K C^E$, and give the BMS replication of the digital payoff $\mathbf{1}_{\{S_T \geq K\}}$.

Solution. Fix $K > 0$ and $\Delta K > 0$. At maturity, define

$$\Pi_T^1 = \Delta K \mathbf{1}_{\{S_T \geq K\}} + (S_T - (K + \Delta K))^+$$

and

$$\Pi_T^2 = \Delta K \mathbf{1}_{\{S_T \geq K + \Delta K\}} + (S_T - (K + \Delta K))^+.$$

Compare these payoffs with $(S_T - K)^+$. If $S_T < K$, all three are zero. If $K \leq S_T < K + \Delta K$, then

$$\Pi_T^2 = 0 \leq S_T - K \leq \Delta K = \Pi_T^1.$$

If $S_T \geq K + \Delta K$, all three equal $S_T - K$. Hence

$$\Pi_T^2 \leq (S_T - K)^+ \leq \Pi_T^1.$$

Absence of arbitrage preserves the order of prices, so for $t < T$,

$$\Delta K C^D(t, S, K + \Delta K) + C^E(t, S, K + \Delta K) \leq C^E(t, S, K)$$

and

$$C^E(t, S, K) \leq \Delta K C^D(t, S, K) + C^E(t, S, K + \Delta K).$$

Equivalently,

$$C^D(t, S, K) \geq \frac{C^E(t, S, K) - C^E(t, S, K + \Delta K)}{\Delta K} \geq C^D(t, S, K + \Delta K).$$

Letting $\Delta K \downarrow 0$, using the assumed continuity of C^D in K and differentiability of C^E , gives

$$C^D(t, S, K) = -\partial_K C^E(t, S, K).$$

In the BMS model,

$$C^E(t, x, K) = xN(d_1) - Ke^{-r\tau}N(d_2), \quad \tau = T - t.$$

Differentiating with respect to K , or pricing directly under $\mathbb{Q}$, gives

$$C^D(t, x, K) = e^{-r\tau}N(d_2), \quad d_2 = \frac{\log(x/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

The replicating initial capital is therefore

$$x_G = C^D(0, S_0, K) = e^{-rT}N(d_2(0, S_0, K)).$$

The risky holding is the delta of the digital price:

$$\theta_t^G = \partial_x C^D(t, S_t, K) = \frac{e^{-r(T-t)}\varphi(d_2(t, S_t, K))}{S_t\sigma\sqrt{T-t}}, \quad 0 \leq t < T.$$

Together with the cash account holding

$$\alpha_t^G = \frac{C^D(t, S_t, K) - \theta_t^G S_t}{S_t^0},$$

this self-financing strategy replicates $\mathbf{1}_{\{S_T \geq K\}}$. □

Exercise II.7 (Digital call, asset-or-nothing call, and replication by calls). **Restatement.** Compute the BMS digital call price and its delta, price the payoff $S_T \mathbf{1}_{\{S_T \geq K\}}$, and decide whether the digital can be replicated using only the money-market account and the European call as risky traded asset.

Solution. Let $\tau = T - t$. The digital call price is

$$C^D(t, x, K) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r\tau} \mathbf{1}_{\{S_T \geq K\}} \mid S_t = x \right] = e^{-r\tau} N(d_2),$$

where

$$d_2 = \frac{\log(x/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

Its sensitivity with respect to the underlying price is

$$\partial_x C^D(t, x, K) = \frac{e^{-r\tau}\varphi(d_2)}{x\sigma\sqrt{\tau}}.$$

For the asset-or-nothing payoff $S_T \mathbf{1}_{\{S_T \geq K\}}$,

$$\mathbb{E}^{\mathbb{Q}} \left[e^{-r\tau} S_T \mathbf{1}_{\{S_T \geq K\}} \mid S_t = x \right] = xN(d_1),$$

with

$$d_1 = d_2 + \sigma\sqrt{\tau} = \frac{\log(x/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

Now suppose the traded risky asset is the European call itself,

$$C^E(t, x, K) = xN(d_1) - Ke^{-r\tau}N(d_2),$$

together with the money-market account. Its delta is

$$\partial_x C^E(t, x, K) = N(d_1) > 0 \quad (t < T, x > 0).$$

The discounted digital and discounted call price processes have Brownian coefficients proportional to their deltas. Therefore choosing

$$\eta_t = \frac{\partial_x C^D(t, S_t, K)}{\partial_x C^E(t, S_t, K)} = \frac{e^{-r(T-t)}\varphi(d_2(t, S_t, K))}{S_t\sigma\sqrt{T-t}N(d_1(t, S_t, K))}$$

units of the European call matches the risky exposure of the digital. The remaining value is invested in the money-market account:

$$\alpha_t = \frac{C^D(t, S_t, K) - \eta_t C^E(t, S_t, K)}{S_t^0}.$$

Thus, for every $t < T$, the digital can be dynamically replicated using only the risk-free asset and the European call as the risky traded asset. The position becomes singular near maturity, as expected for a discontinuous payoff. $\square$

Exercise II.8 (Compound call). **Restatement.** Let $0 < T_1 < T$. The payoff at T_1 is a call on the time- T_1 price of a European call maturing at T :

$$G = (C_{T_1}(S, T, K) - K_1)^+.$$

Derive the price, express it with a bivariate normal probability, and explain why it is replicable using only S and the risk-free asset.

Solution. Let $u(t, x) = C^E(t, x, T, K)$ be the usual BMS call price on S with maturity T and strike K . For $t \leq T_1$, put

$$\tau = T_1 - t, \quad \tau_1 = T - T_1.$$

By risk-neutral pricing,

$$p_t = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T_1-t)} (u(T_1, S_{T_1}) - K_1)^+ \mid \mathcal{F}_t \right].$$

Conditionally on $S_t = x$,

$$S_{T_1} = x \exp \left(\left(r - \frac{1}{2}\sigma^2 \right) \tau + \sigma\sqrt{\tau} Z \right), \quad Z \sim \mathcal{N}(0, 1).$$

Hence $p_t = v(T_1 - t, S_t)$, where

$$v(\tau, x) = e^{-r\tau} \mathbb{E} \left[\left(u \left(T_1, x e^{(r - \frac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z} \right) - K_1 \right)^+ \right].$$

The function $x \mapsto u(T_1, x)$ is continuous, strictly increasing, tends to 0 as $x \downarrow 0$, and tends to $+\infty$ as $x \rightarrow \infty$. Therefore there is a unique $x_1 > 0$ such that

$$u(T_1, x_1) = K_1.$$

We use the sign convention

$$d = \frac{\log(x/x_1) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}},$$

so that

$$xe^{(r-\frac{1}{2}\sigma^2)\tau+\sigma\sqrt{\tau}Z} > x_1 \iff Z > -d.$$

Thus

$$v(\tau, x) = e^{-r\tau} \mathbb{E} \left[u \left(T_1, xe^{(r-\frac{1}{2}\sigma^2)\tau+\sigma\sqrt{\tau}Z} \right) \mathbf{1}_{\{Z > -d\}} \right] - e^{-r\tau} K_1 N(d).$$

To expand the remaining expectation, introduce an independent $Z_1 \sim \mathcal{N}(0, 1)$. Conditional on S_{T_1} , the time- T stock is

$$S_T = x \exp \left(\left(r - \frac{1}{2}\sigma^2 \right) (\tau + \tau_1) + \sigma(\sqrt{\tau}Z + \sqrt{\tau_1}Z_1) \right).$$

Using the call pricing formula inside $u(T_1, \cdot)$ gives

$$v(\tau, x) + e^{-r\tau} K_1 N(d) = \mathbb{E} \left[\left(xe^{\sigma(\sqrt{\tau}Z + \sqrt{\tau_1}Z_1) - \frac{1}{2}\sigma^2(\tau + \tau_1)} - K e^{-r(\tau + \tau_1)} \right) \mathbf{1}_A \right],$$

where

$$A = \left\{ \sigma(\sqrt{\tau}Z + \sqrt{\tau_1}Z_1) > - \left(\log(x/K) + \left(r - \frac{1}{2}\sigma^2 \right) (\tau + \tau_1) \right) \right\} \cap \{Z > -d\}.$$

Now use the notation from the statement

$$N_2(y, y_1, \rho) = \mathbb{P}(Z < y, Z + \rho Z_1 < y_1), \quad \rho = \sqrt{\frac{\tau_1}{\tau}}.$$

Define

$$a_1 = \frac{\log(x/K) + (r + \frac{1}{2}\sigma^2)(\tau + \tau_1)}{\sigma\sqrt{\tau}}, \quad a_2 = \frac{\log(x/K) + (r - \frac{1}{2}\sigma^2)(\tau + \tau_1)}{\sigma\sqrt{\tau}}.$$

By symmetry of (Z, Z_1) ,

$$\mathbb{P}(A) = N_2(d, a_2, \rho).$$

Completing the square in the exponential term shifts (Z, Z_1) by $(\sigma\sqrt{\tau}, \sigma\sqrt{\tau_1})$, which gives

$$\mathbb{E} \left[e^{\sigma(\sqrt{\tau}Z + \sqrt{\tau_1}Z_1)} \mathbf{1}_A \right] = e^{\frac{1}{2}\sigma^2(\tau + \tau_1)} N_2(d + \sigma\sqrt{\tau}, a_1, \rho).$$

Therefore

$$\boxed{v(\tau, x) = x N_2(d + \sigma\sqrt{\tau}, a_1, \rho) - K e^{-r(\tau + \tau_1)} N_2(d, a_2, \rho) - K_1 e^{-r\tau} N(d)}.$$

Taking $\tau = T_1 - t$ gives the time- t price $p_t = v(T_1 - t, S_t)$.

Finally, the payoff is a measurable function of S_{T_1} :

$$G = (u(T_1, S_{T_1}) - K_1)^+.$$

It is square-integrable in the lognormal model. By the completeness result behind Poly, Prop. 3.3, or equivalently by applying the BMS PDE to $(t, x) \mapsto v(T_1 - t, x)$ on $t < T_1$, it admits a self-financing replication with the money-market account and the stock S only. The risky holding is the delta $\partial_x v(T_1 - t, S_t)$. $\square$